QR-BASED XEROX AUTOMATION

Core Concept & System Overview

Purpose: Make document printing simple for customers and easier to manage for the Xerox owner. A single daily token connects payment, documents, print jobs, progress and collection.

1. Customer experience

- Customer scans the shop's QR code and opens the website.
- No registration, login, name, phone number or OTP is required.
- A temporary browser session keeps the upload and print settings together while the customer is using the website. It is not a permanent customer identity or the order ID.
- Customer uploads one or more PDF documents.
- Customer selects printing preferences: colour or black and white, single- or double-sided, page ranges and number of copies.
- Customer reviews the quotation and completes payment.

2. Document verification

- The system checks that each uploaded PDF is valid and readable.
- It extracts relevant details such as page count and file size.
- It validates the requested print settings and calculates the price.
- The customer sees the final quotation before paying.
- Only after payment is verified does the system release the order for printing.

3. Order and daily token

- Every successfully paid order receives a customer-facing token, for example XR-101.
- The token is the order reference shown to the customer and used by the owner to find the order.
- All documents and print jobs for that customer order are linked to the same token.
- Tokens increase sequentially: XR-101, XR-102, XR-103, and so on.
- The daily counter resets at midnight India time and starts again at XR-101.
- Because tokens repeat on different days, the backend distinguishes orders using the shop and business date as well as the token.

4. Documents and print jobs

- One order can contain several documents, and each document can have its own print settings and print job.
- Example: Order XR-101 contains Assignment.pdf → Print Job 1; Notes.pdf → Print Job 2; Project.pdf → Print Job 3.

5. Printing and completion verification

- A local printing agent on the shop's Windows computer connects the backend to the Xerox printer.
- Orders are processed sequentially: the next customer's order waits until the current order's physical stack has been separated.

- The system tracks print-job states and printer/spooler information where supported.
- A print command being submitted or disappearing from the spooler is not, by itself, proof that the pages physically came out correctly.
- The owner confirms that the completed stack has been removed and placed in the collection area before the next order is released.

6. Customer tracking

- After payment, the customer sees the token number, such as XR-101.
- The tracking view can show payment status, order status, number of documents, printing progress and estimated waiting time.
- Possible statuses include Queued, Printing and Ready for collection.
- Estimated waiting time is calculated from the current queue and measured printer performance; it is an estimate, not a guarantee.
- The customer can revisit tracking through a secure QR tracking ticket without logging in. The sequential token alone should not grant access to private documents.

7. Xerox owner dashboard

- Today's token list: XR-101, XR-102, XR-103, and so on.
- Total orders, pending orders and the order currently being printed.
- Upcoming printing queue and estimated waiting time.
- Documents and individual print settings under each token.
- Payment status, print-job progress and collection status.
- Search for an order using its token.

8. Daily data management

- Temporary browser sessions and unnecessary temporary files can be cleared automatically.
- Completed and collected orders and their documents can be deleted according to the configured retention policy.
- Unfinished paid orders should remain available until they are resolved; otherwise a midnight deletion could lose a paid customer's uncollected print job.
- The daily token counter resets at midnight India time.
- Keep the minimum payment records needed for reconciliation, refunds, disputes and applicable accounting requirements.

9. Core concept at a glance

ONE ORDER → ONE TOKEN → MULTIPLE DOCUMENTS → MULTIPLE PRINT JOBS → ONE PHYSICAL STACK → ONE COLLECTION

The customer remembers one simple token. The owner uses that same token to locate and manage the order. The backend coordinates payment verification, document handling, printing progress, waiting-time estimates and collection.